Jan Hofmann

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Education

Stony Brook University — **Stony Brook, New York**

Ph.D. in Chemistry — *Expected Q2 2027*
GPA: 3.88/4.00 (US grading scale)

Leipzig University — **Leipzig, Germany**

M.Sc. in Chemistry — *December 2021*
Final Grade: 1.3

Leipzig University — **Leipzig, Germany**

B.Sc. in Chemistry — *February 2018*
Final Grade: 1.9

Skills

Quantitative Analysis: Microsoft Excel, Python (pandas, SciPy, scikit-learn), statistical modeling, multivariate analysis (NMF, PCA).

Project Coordination & Communication: Experimental planning, international research collaboration, Microsoft PowerPoint presentations, German (native), English (fluent).

Programming and Automation: Python data-processing pipelines, batch analysis, data visualization, structured result export.

Professional Experience

Stony Brook University, Department of Chemistry — **Stony Brook, NY**

Graduate Researcher — *August 2021 - Present*

- Applied statistical modeling and machine learning (NMF, PCA) to complex experimental datasets, identifying previously unresolved structural changes in framework materials.
- Led international synchrotron experiments and coordinated data collection and analysis across institutions, contributing to 13 peer-reviewed publications, including first-author work.
- Standardized experimental measurements through instrument calibration, automated acquisition workflows, and SOP development.
- Quantified a sixfold greater negative thermal expansion response than benchmark materials and developed a geometric model linking local structural changes to macroscopic behavior, identifying preferential alignment as the key mechanism.

Leibniz Institute of Surface Engineering (IOM) — **Leipzig, Germany**

Research Assistant — *July 2019 - October 2021*

- Analyzed high-entropy alloy catalysts, identifying phase segregation and structural changes following thermal treatment.
- Applied equivalent-circuit models to electrochemical impedance spectra, assessing charge-transfer resistance and diffusion limitations in fuel-cell electrodes.

Cardiff Catalysis Institute **Cardiff, United Kingdom**

Student Researcher *September 2018 - April 2019*

- Synthesized supported catalysts using wet impregnation and sol immobilization to compare preparation methods.
- Characterized catalyst morphology and elemental composition using scanning electron microscopy.
- Quantified activity and selectivity of bimetallic catalysts using gas chromatography.

TOTAL Refinery Mitteldeutschland **Leuna, Germany**

R&D Intern *January 2018 - July 2018*

- Established a quantitative correlation between fuel oxidation stability testing methods, demonstrating 20-fold shorter induction times and 90% lower sample consumption.
- Authored SOPs for sample preparation and operation of Anton Paar PetroOxy instruments, supporting the implementation of a new analysis method at the refinery.
- Evaluated the influence of antioxidant additives on the resistance of gasoline to oxidative degradation.

Leadership & Volunteer Experience

Friends of the Chemistry Olympiad e.V. **Leipzig, Germany**

Organizer *August 2014 - Present*

- Organized state-level rounds of the annual chemistry competition.
- Managed speaker invitations and outreach to local newspapers to promote events.
- Created exam questions for the competition.

Graduate Student Organization, Stony Brook University **Stony Brook, NY**

Senator *August 2024 - August 2026*

- Coordinated $2,000 in annual funding for initiatives focused on improving graduate student life at Stony Brook.
- Assisted students with conference funding applications through one-on-one consultations and group workshops.
- Planned graduate student events in collaboration with Stony Brook student clubs.